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SET-I

ODD END SEMESTER EXAMINATION, DECEMBER, 2025

Name of the Program: B. Tech Biotechnology

Semester: III

Name of the Course: Cell Biology

Course Code: TBT302

Time: 3 hours

Maximum Marks: 100

Note:

- i. All questions are compulsory.
- ii. Answer any two sub questions among a, b & c in each main question
- iii. Total marks in each main question are twenty.
- iv. Each question carries 10 marks

Q1	20 Marks	COs
(a) Differentiate between apoptosis and necrosis at the molecular and cellular levels. Explain the intrinsic and extrinsic pathways of apoptosis, describing the role of caspases and Bcl-2 family proteins.		CO4, CO3
(b) Define the cell cycle and describe its major phases. Explain the significance of each phase of cell cycle in detail.		
(c) Define cell signaling and explain its importance in cellular communication and coordination. Describe the types of cell signaling. Discuss the general steps involved in signal reception, transduction, and response.		
Q2	20 Marks	
(a) Define cancer and explain its fundamental characteristics. Discuss the different types of cancers and describe their causes, stages, and modes of spread (metastasis). Elaborate on common prevention strategies and available treatment approaches of cancer.		CO1, CO3, CO4
(b) Describe the cell theory and explain its historical development. Discuss the contributions of Schleiden, Schwann, and Virchow, and outline the principles of the modern cell theory.		
(c) Explain the JAK-STAT signaling pathway involved in cytokine-mediated communication. Describe their roles in transmitting extracellular signals to the nucleus.		
Q3	20 Marks	

(a)	Explain the roles of cyclins and cyclin-dependent kinases (CDKs) in regulating cell cycle progression.	CO2, CO4
(b)	Compare the structure and functions of mitochondria and chloroplasts. Explain their role in energy metabolism (ATP synthesis) and discuss how they support the endosymbiotic theory of cell evolution.	
(c)	Explain the process of DNA packaging into chromosomes. Include a labeled diagram to illustrate the hierarchical organization of DNA.	
Q4	20 Marks	
(a)	Discuss in detail the roles of the Rb and p53 tumor suppressor genes in regulating the cell cycle and preventing uncontrolled cell proliferation. Explain how mutations or inactivation of these genes contribute to cancer development.	CO3, CO4, CO5
(b)	Explain the mechanisms of cell-cell interactions and describe the different types of cell junctions in detail.	
(c)	Define cell counting and explain its importance in cell biology and biotechnology experiments. Describe in detail about the hemocytometer technique, including its principle, procedure, calculation formula, and precautions to be taken for accurate results	
Q5	20 Marks	
(a)	Explain the characteristics and importance of embryonic stem cells (ESCs) in developmental biology and regenerative medicine. Discuss ethical issues associated with their use.	CO2, CO4
(b)	Differentiate between numerical and structural chromosomal abnormalities, explaining how they arise during cell division	
(c)	Describe the transport of solutes across the cell membrane. Explain the differences between passive transport and active transport.	